PS2 Predictive Analytics

Barni

2.9

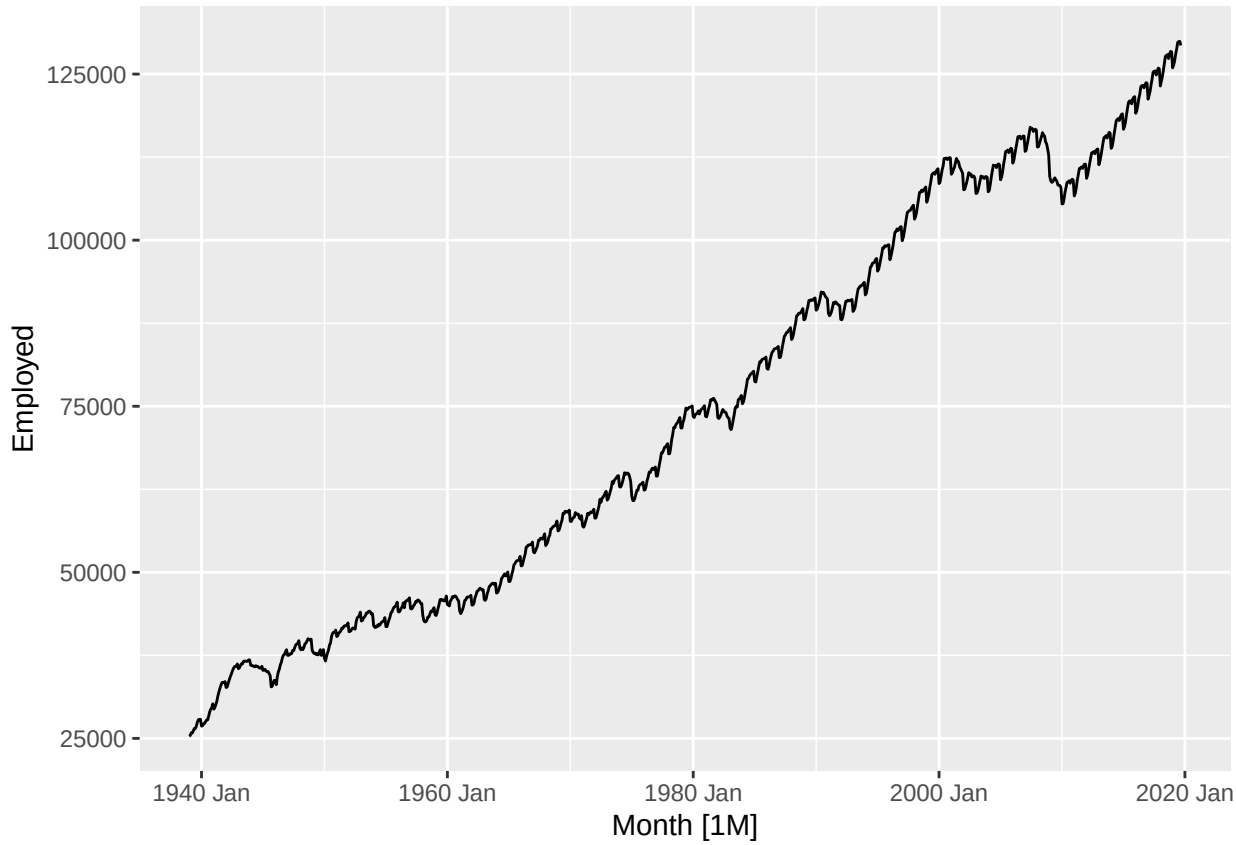
Use the following graphics functions: autoplot(), gg_season(), gg_subseries(), gg_lag(), ACF() and explore features from the following time series: "Total Private" Employed from *us_employment*, Bricks from *aus_production*, Hare from *pelt*, "H02" Cost from *PBS*, and [Barrels from] *us_gasoline*.

- Can you spot any seasonality, cyclicity and trend?
- What do you learn about the series?
- What can you say about the seasonal patterns?
- Can you identify any unusual years?

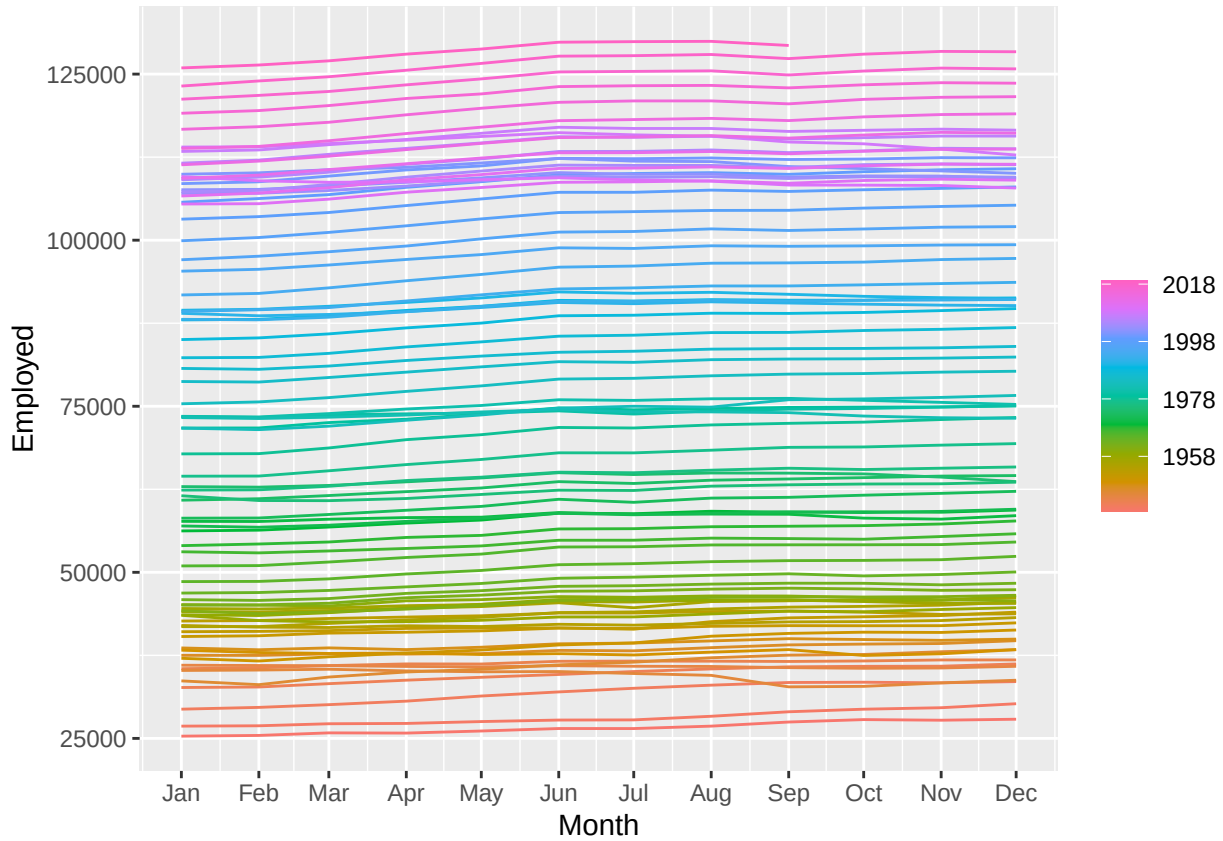
```r
# Load libraries:
library(fpp3) # here we only need fpp3 the library of the book that includes all the tidyverse family.
```

```
## -- Attaching packages ---------------------------------------------- fpp3 0.5 --

## v tibble      3.2.1     v tsibble     1.1.4
## v dplyr       1.1.4     v tsibbledata 0.4.1
## v tidyr       1.3.0     v feasts      0.3.1
## v lubridate   1.9.3     v fable       0.3.3
## v ggplot2     3.4.4     v fabletools  0.3.4

## -- Conflicts ------------------------------------------------- fpp3_conflicts --
## x lubridate::date()    masks base::date()
## x dplyr::filter()      masks stats::filter()
## x tsibble::intersect() masks base::intersect()
## x tsibble::interval()  masks lubridate::interval()
## x dplyr::lag()         masks stats::lag()
## x tsibble::setdiff()   masks base::setdiff()
## x tsibble::union()     masks base::union()
```

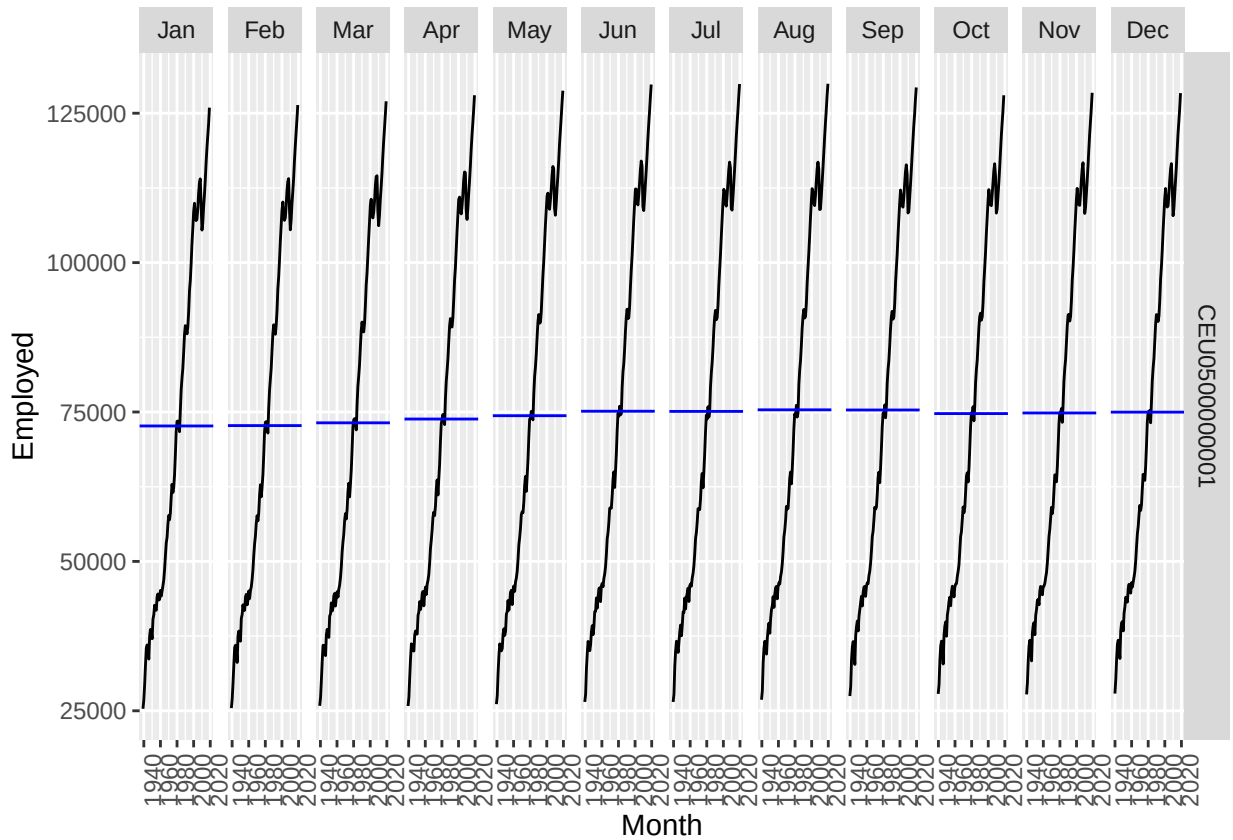
```r
#Total Private Employment in the US
us_employment %>%
  filter(Title == "Total Private") %>%
  autoplot(Employed)
```



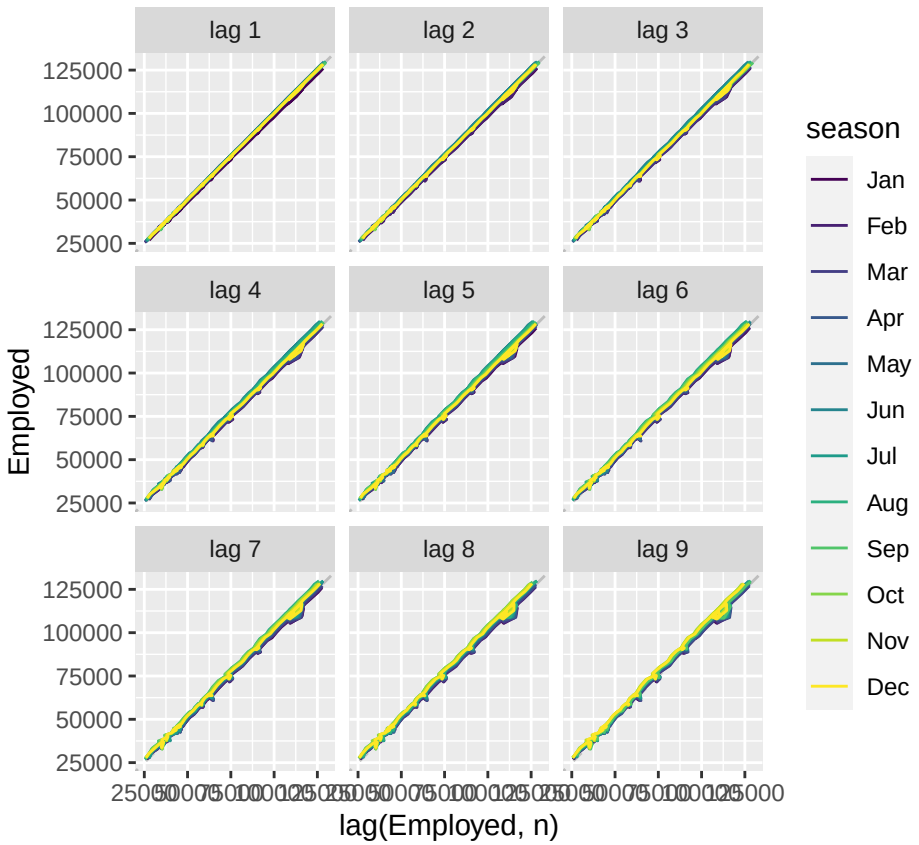

```r
# There is a strong trend and seasonality. Some cyclic behaviour is seen, with
# a big drop due to the global financial crisis.

us_employment %>%
  filter(Title == "Total Private") %>%
  gg_season(Employed)
```



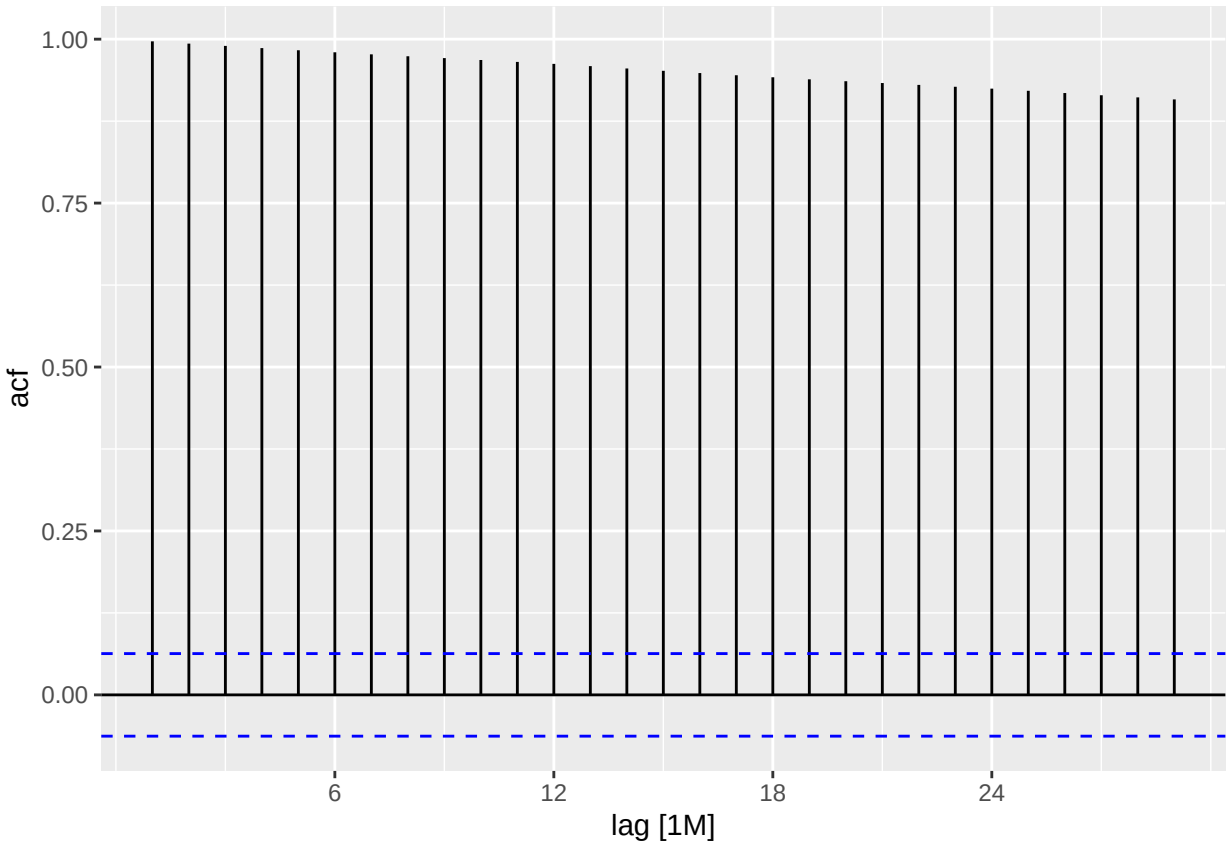

```
us_employment %>%
  filter(Title == "Total Private") %>%
  gg_subseries(Employed)
```




```
us_employment %>%
  filter(Title == "Total Private") %>%
  gg_lag(Employed)
```



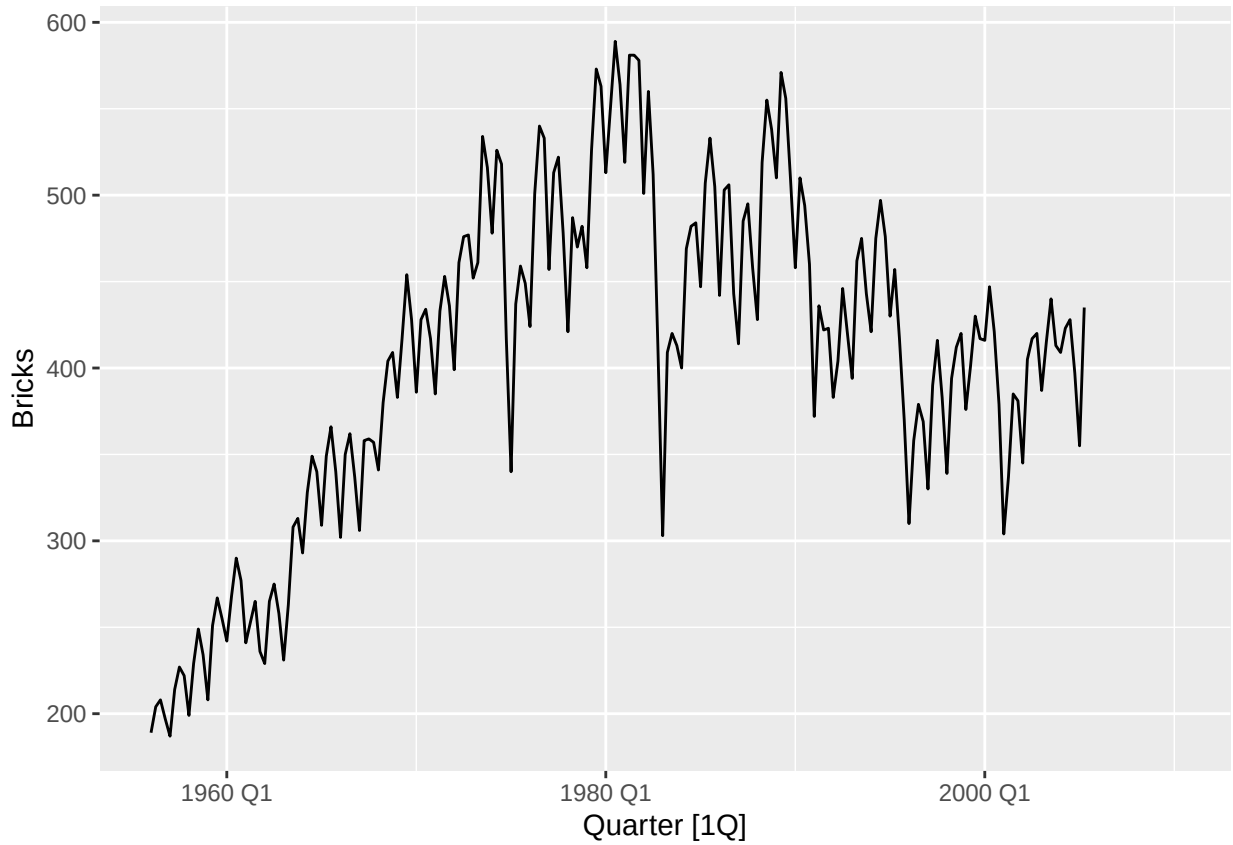

```
us_employment %>%
  filter(Title == "Total Private") %>%
  ACF(Employed) %>%
  autoplot()
```




```
# In all of these plots, the trend is so dominant that it is hard to see
# anything else. We need to remove the trend so we can explore the other
# features of the data.
```

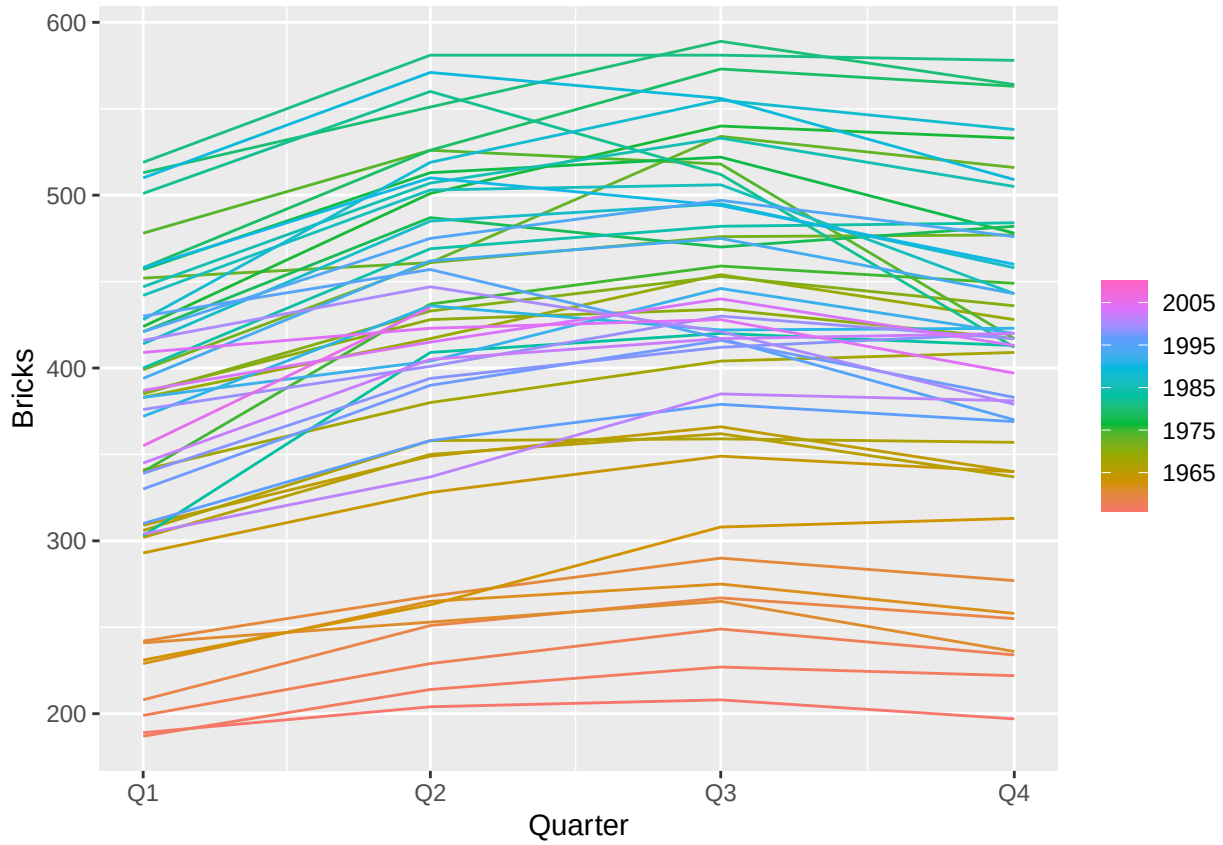
```
# Brick production in Australia

aus_production %>%
  autoplot(Bricks)
```

```
## Warning: Removed 20 rows containing missing values (`geom_line()`).
```



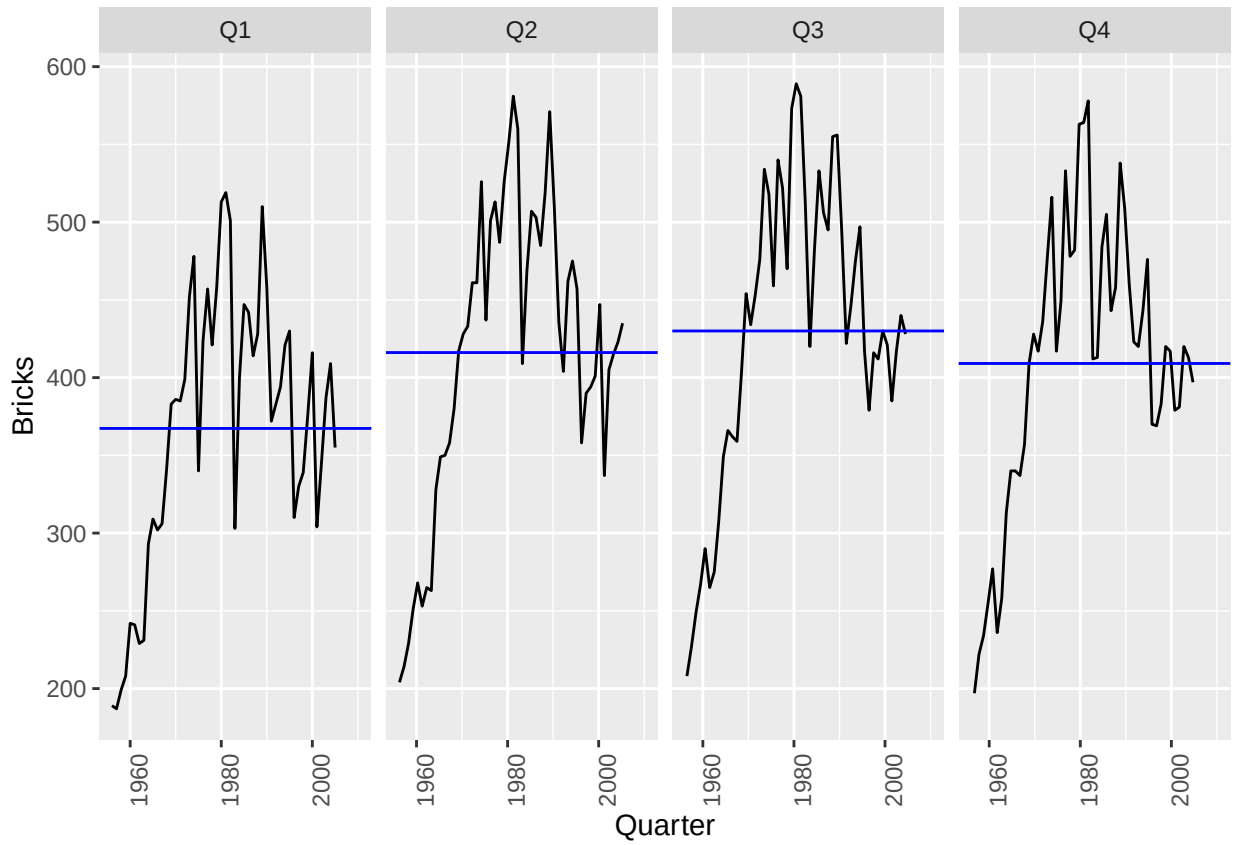

```
# A positive trend in the first 20 years, and a negative trend in the next 25 years.
# Strong quarterly seasonality, with some cyclicity - note the recessions in the
# 1970s and 1980s.

aus_production %>%
  gg_season(Bricks)
```

```
## Warning: Removed 20 rows containing missing values (`geom_line()`).
```



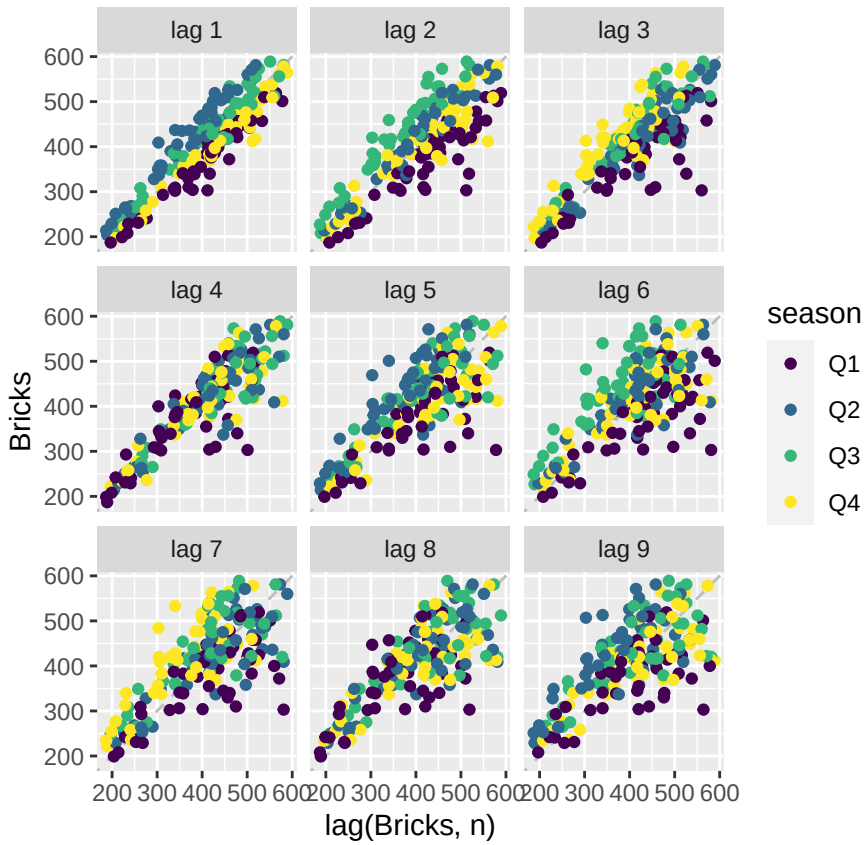

```r
# Brick production tends to be lowest in the first quarter and peak in either
# quarter 2 or quarter 3.

aus_production %>%
  gg_subseries(Bricks)
```

```
## Warning: Removed 5 rows containing missing values (`geom_line()`).
```

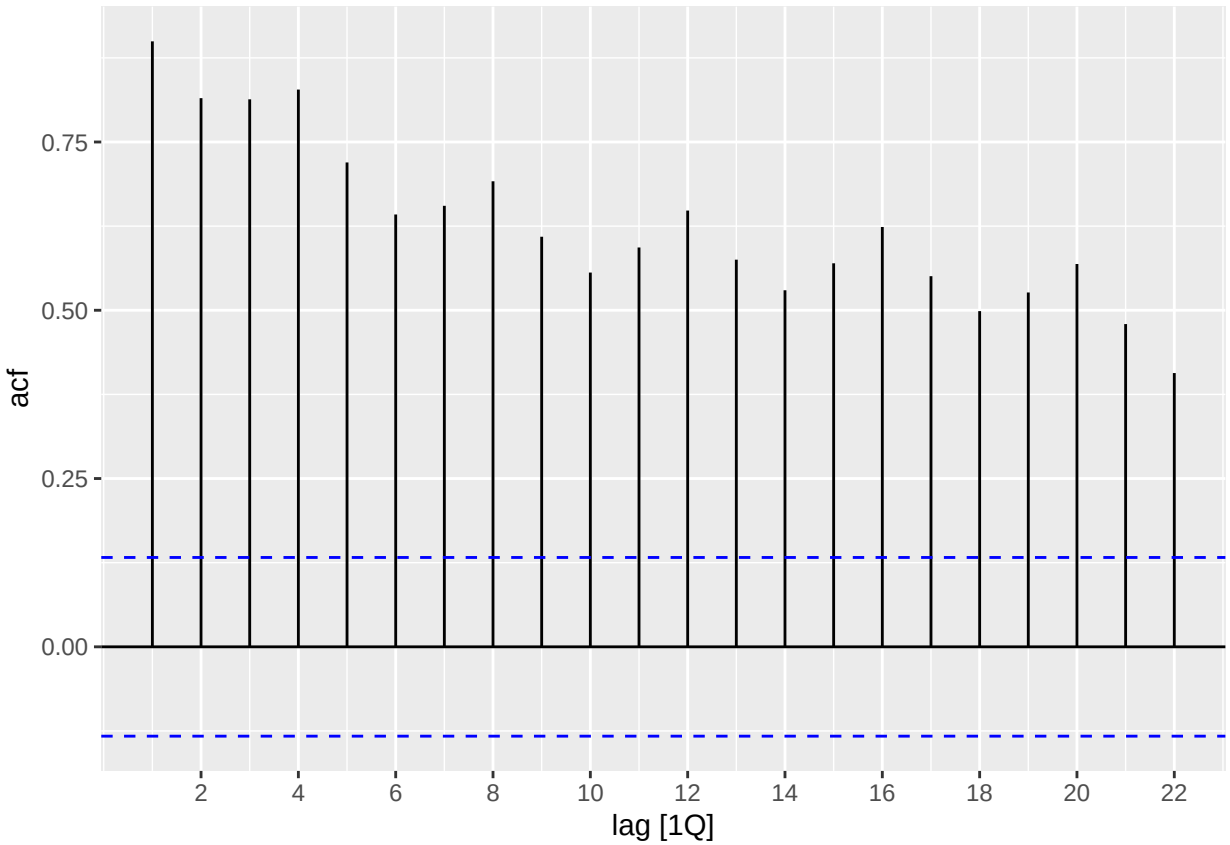



```
# The decrease in the last 25 years has been weakest in Q1.

aus_production %>%
  gg_lag(Bricks, geom='point')
```

```
## Warning: Removed 20 rows containing missing values (gg_lag).
```

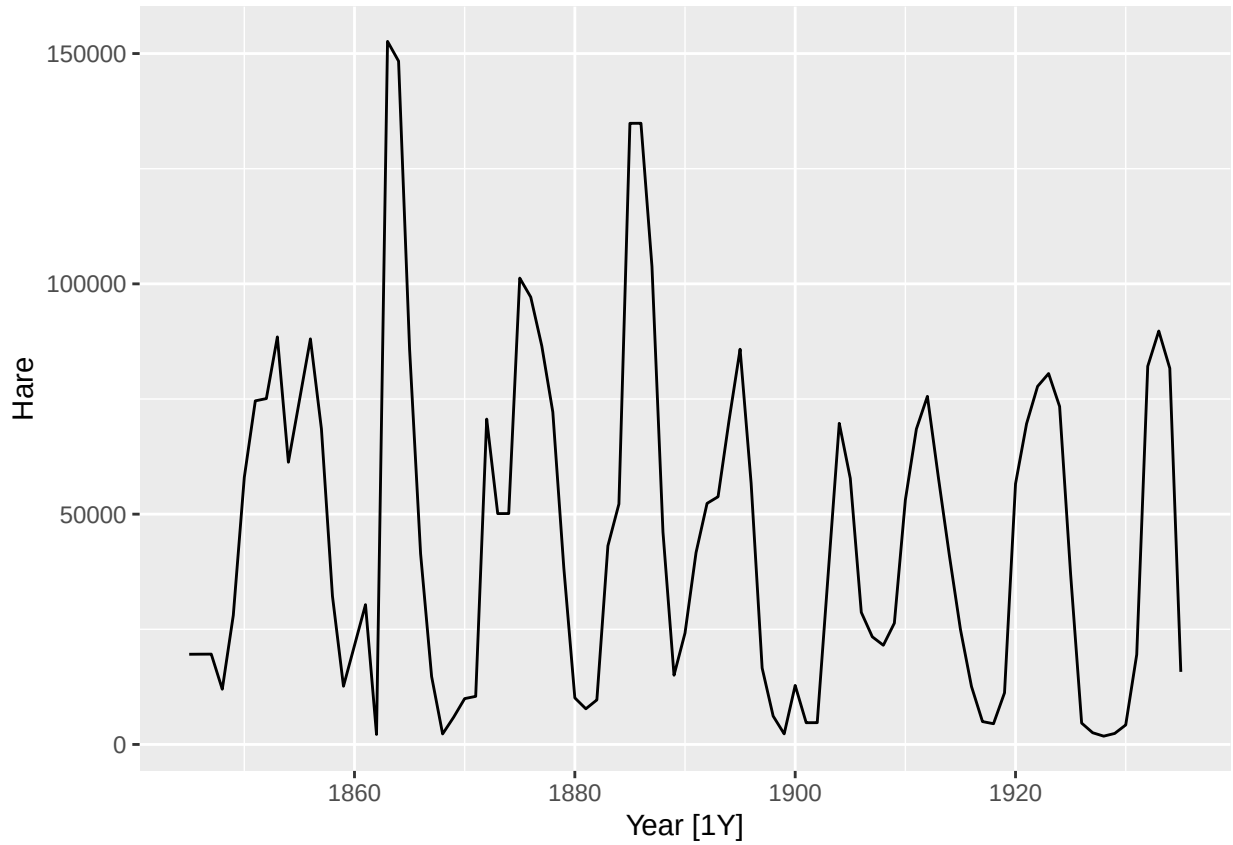



```
aus_production %>%
  ACF(Bricks) %>% autoplot()
```

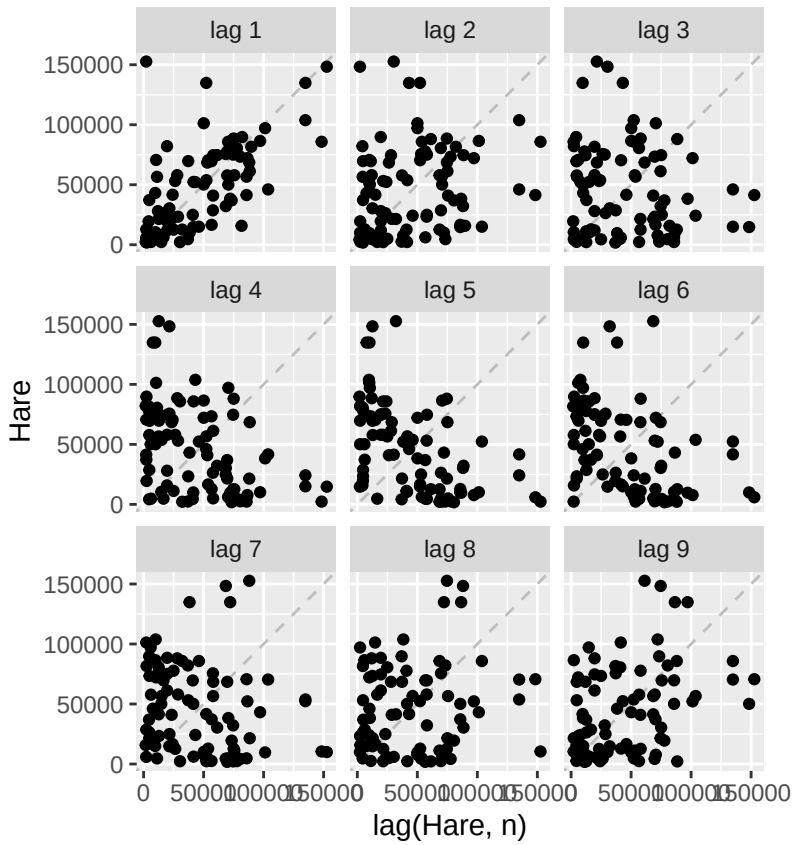



```
# The seasonality shows up as peaks at lags 4, 8, 12, 16, 20, .... The trend is
# seen with the slow decline on the positive side.
```

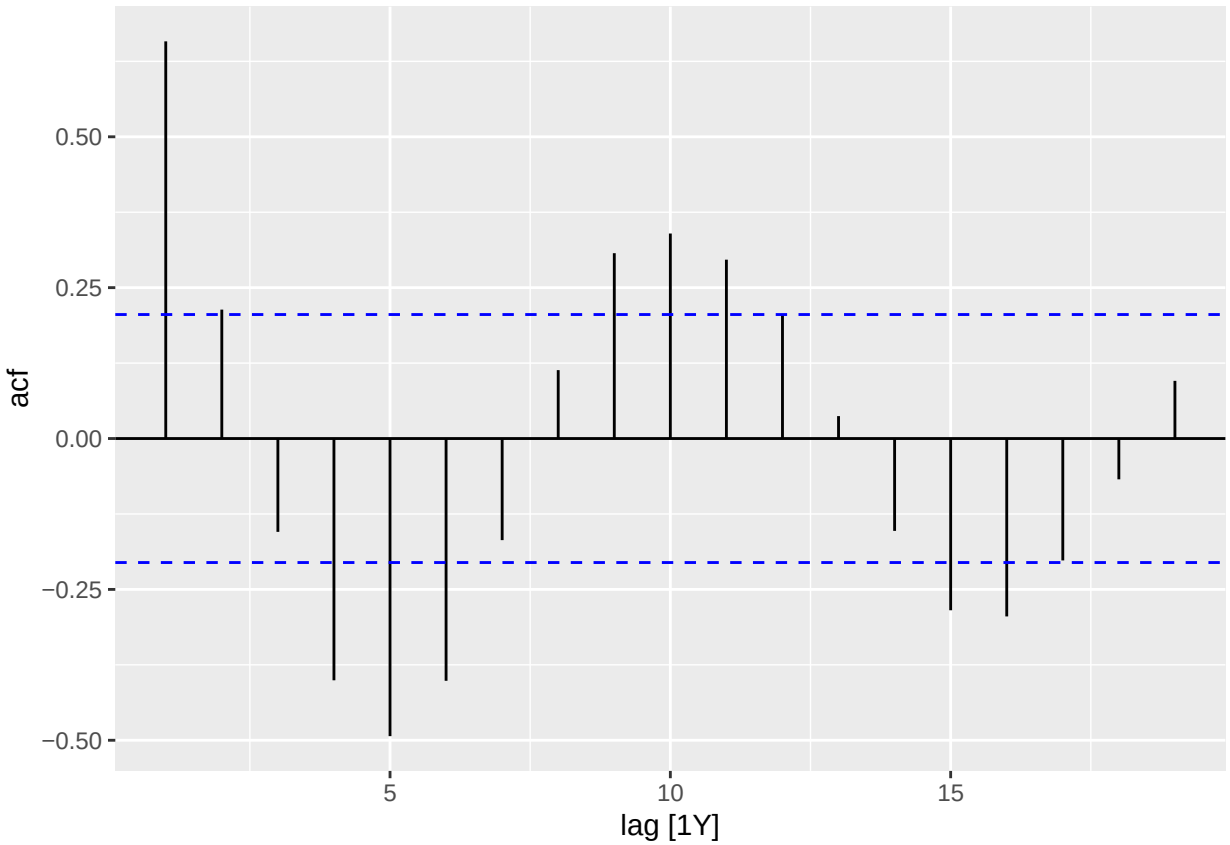
```
# Snow hare trappings in Canada

pelt %>%
  autoplot(Hare)
```




```
# There is some cyclic behaviour with substantial variation in the length of the period.

pelt %>%
  gg_lag(Hare, geom='point')
```

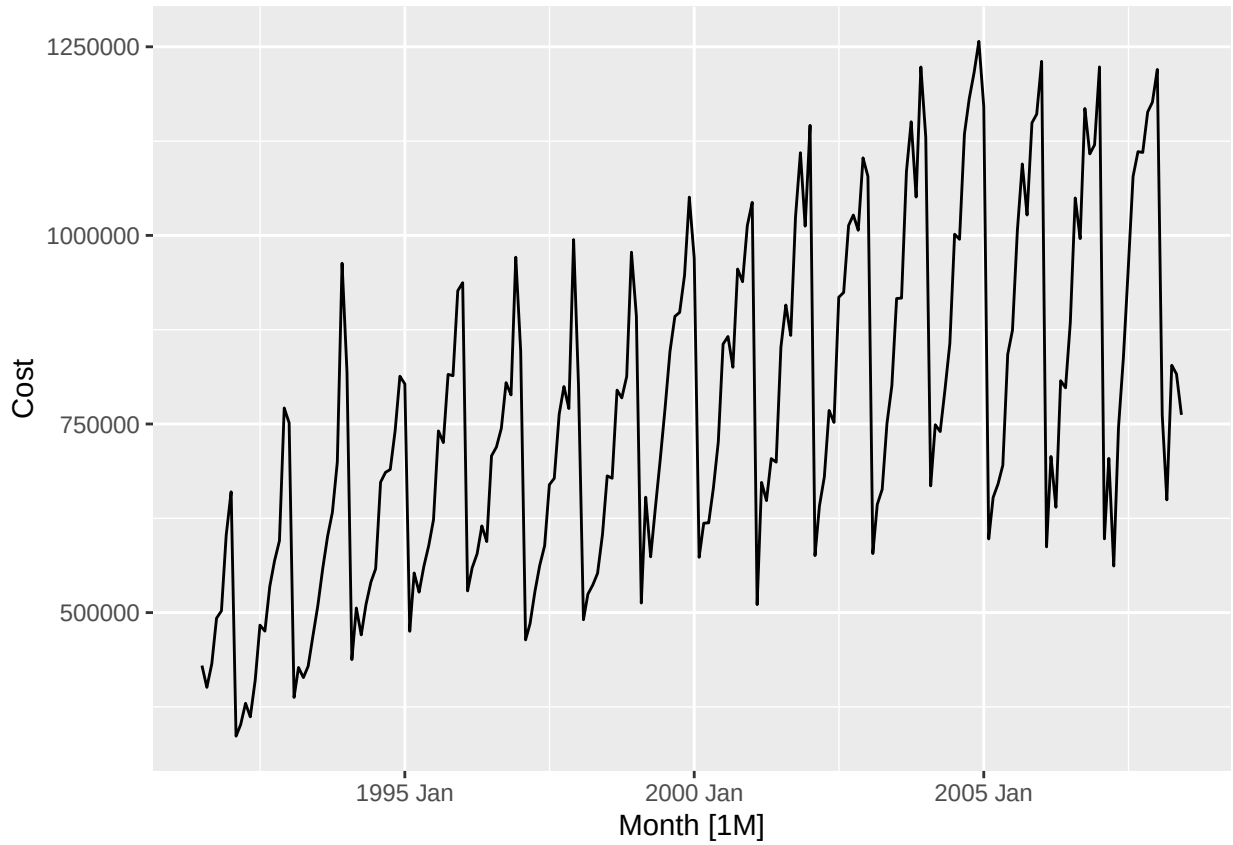



```
pelt %>%
  ACF(Hare) %>% autoplot()
```

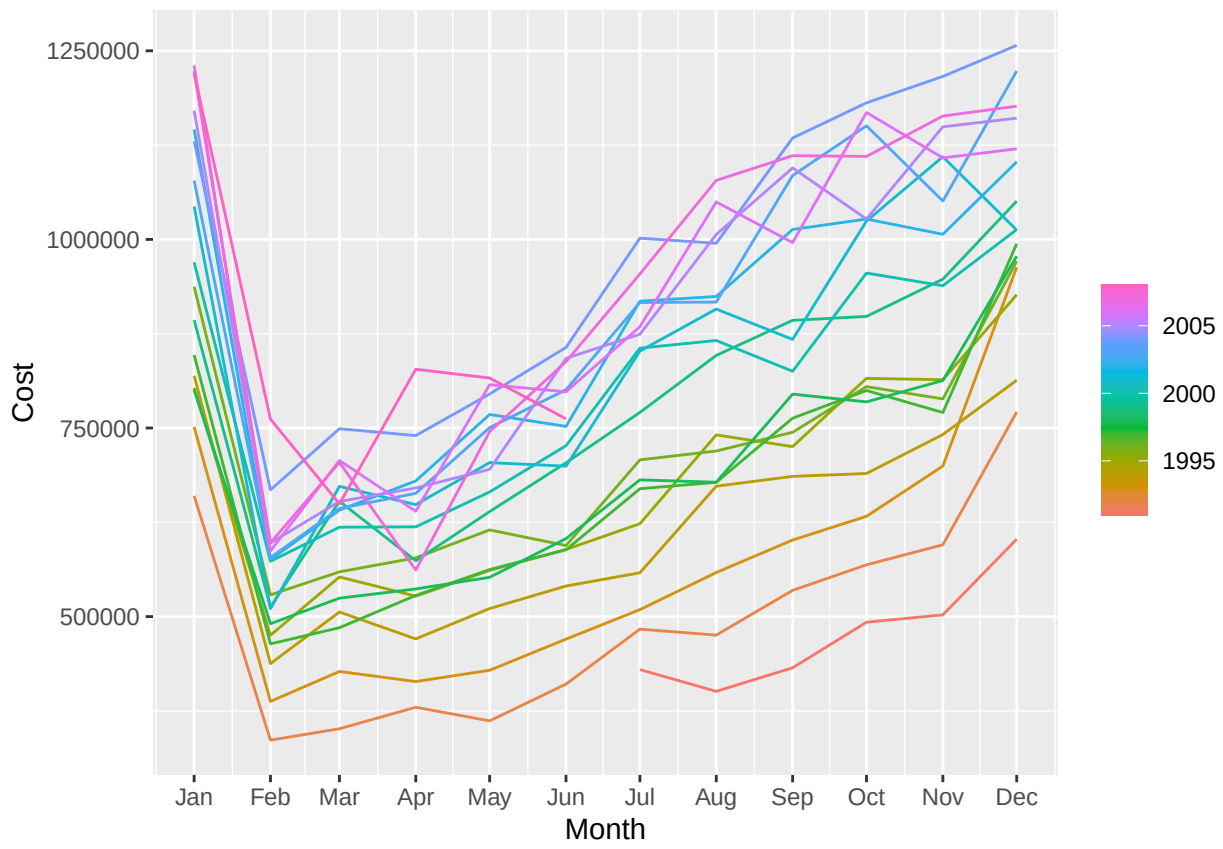



```
# The cyclic period seems to have an average of about 10 (due to the local maximum in ACF at lag 10).
```

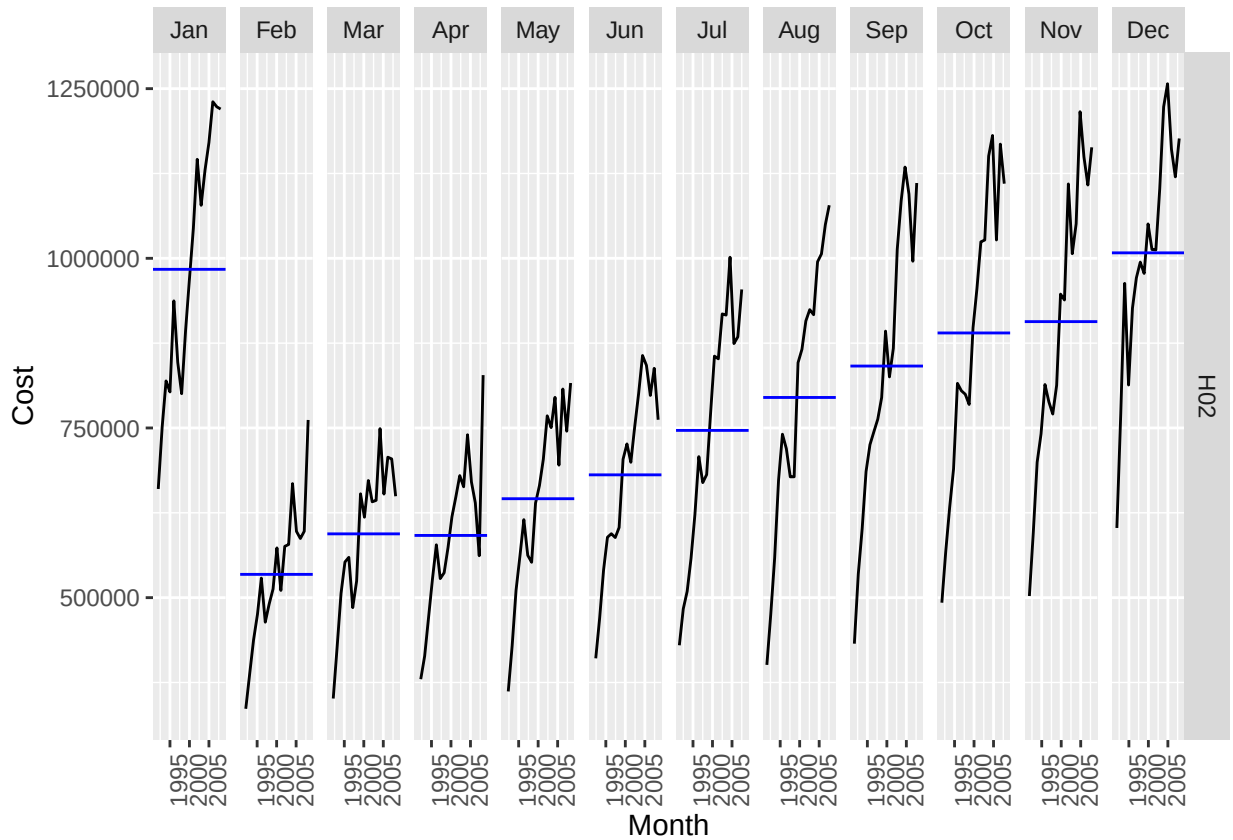
```
#H02 sales in Australia
# There are four series corresponding to H02 sales, so we will add them together.

h02 <- PBS %>%
  filter(ATC2 == "H02") %>%
  group_by(ATC2) %>%
  summarise(Cost = sum(Cost)) %>%
  ungroup()

h02 %>%
  autoplot(Cost)
```



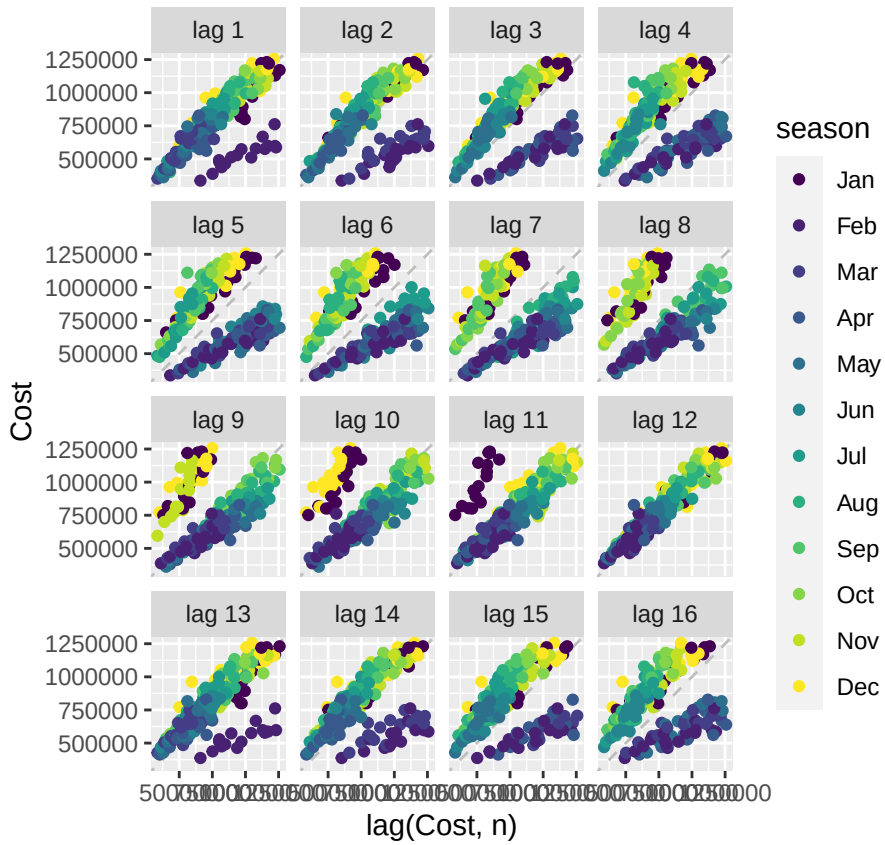

```
# A positive trend with strong monthly seasonality, dropping suddenly every February.

h02 %>%
  gg_season(Cost)
```



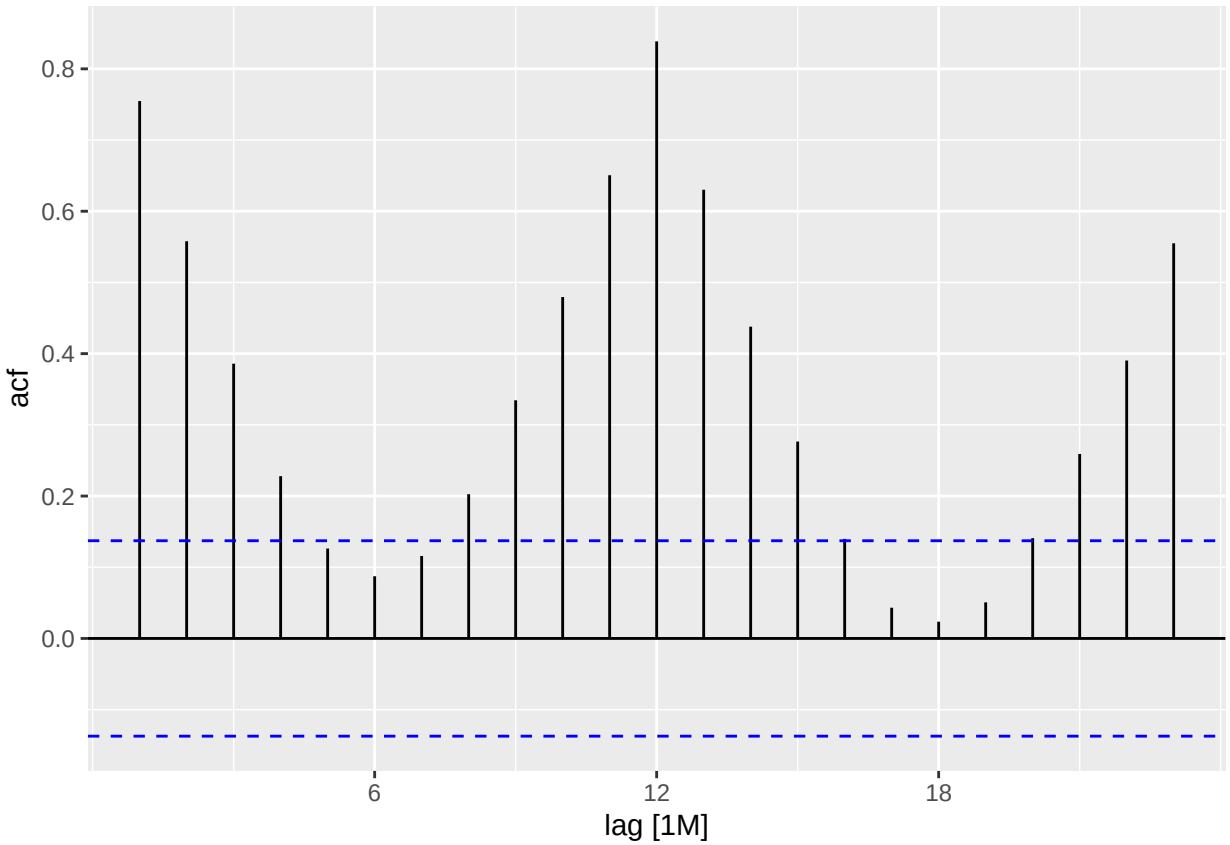

```
h02 %>%
  gg_subseries(Cost)
```




```
# The trends have been greater in the higher peaking months - this leads to
# increasing seasonal variation.

h02 %>%
  gg_lag(Cost, geom='point', lags=1:16)
```

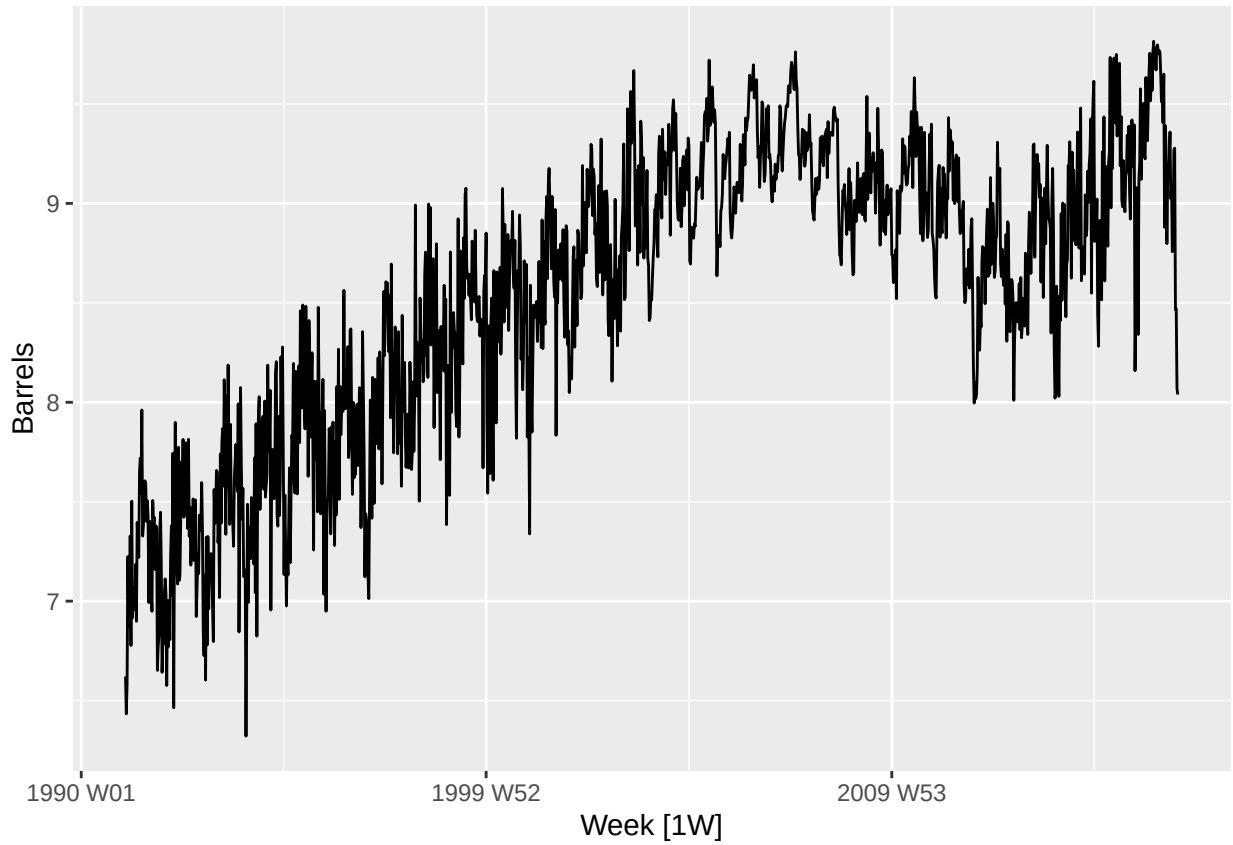



```
h02 %>%
  ACF(Cost) %>% autoplot()
```

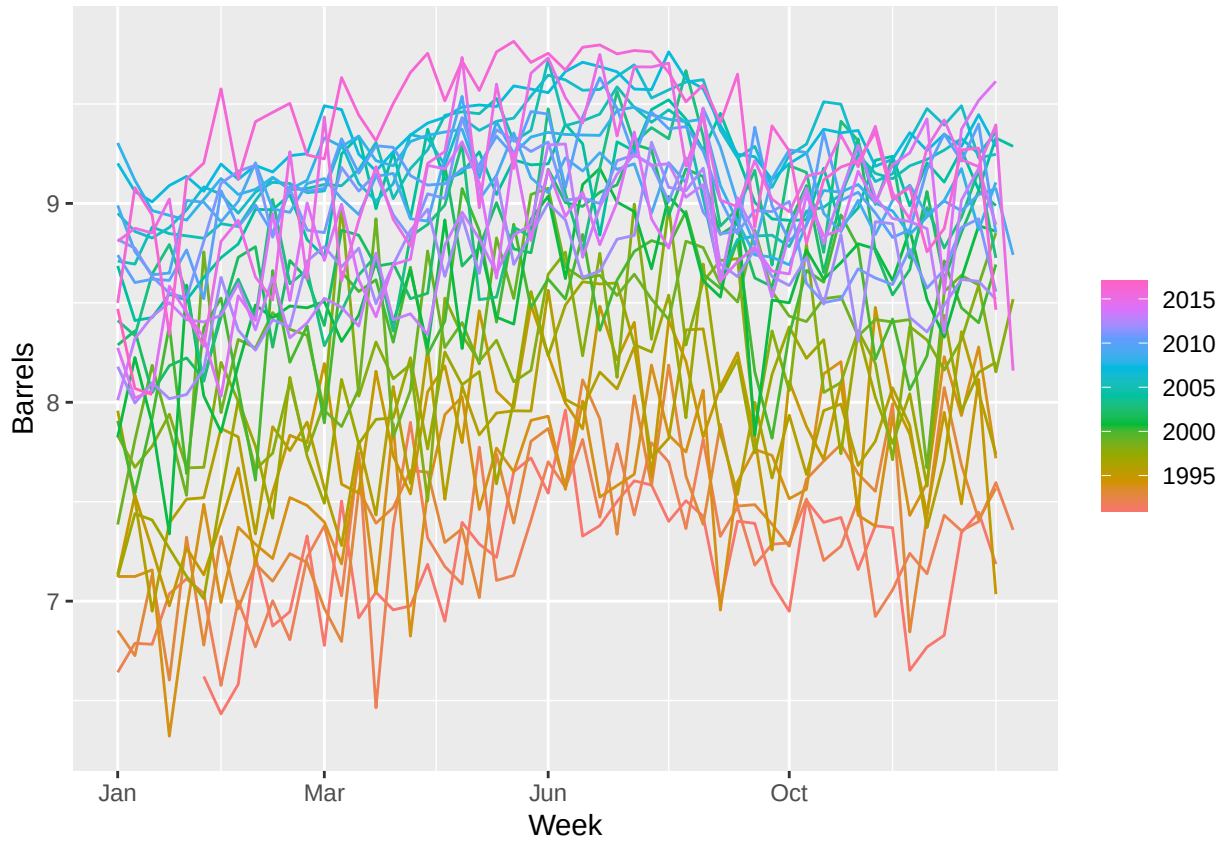



```
# The large January sales show up as a separate cluster of points in the lag plots.
# The strong seasonality is clear in the ACF plot.
```

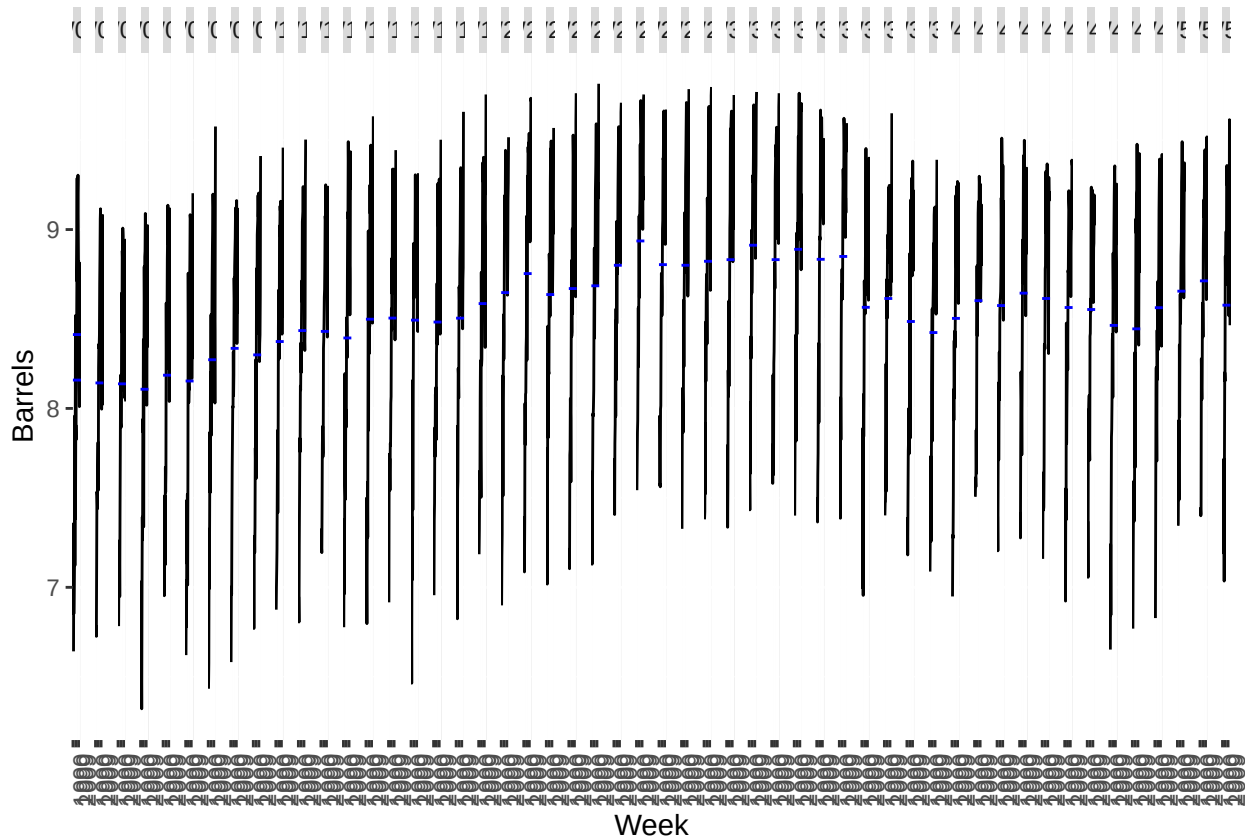
```
# US gasoline sales

us_gasoline %>%
  autoplot(Barrels)
```



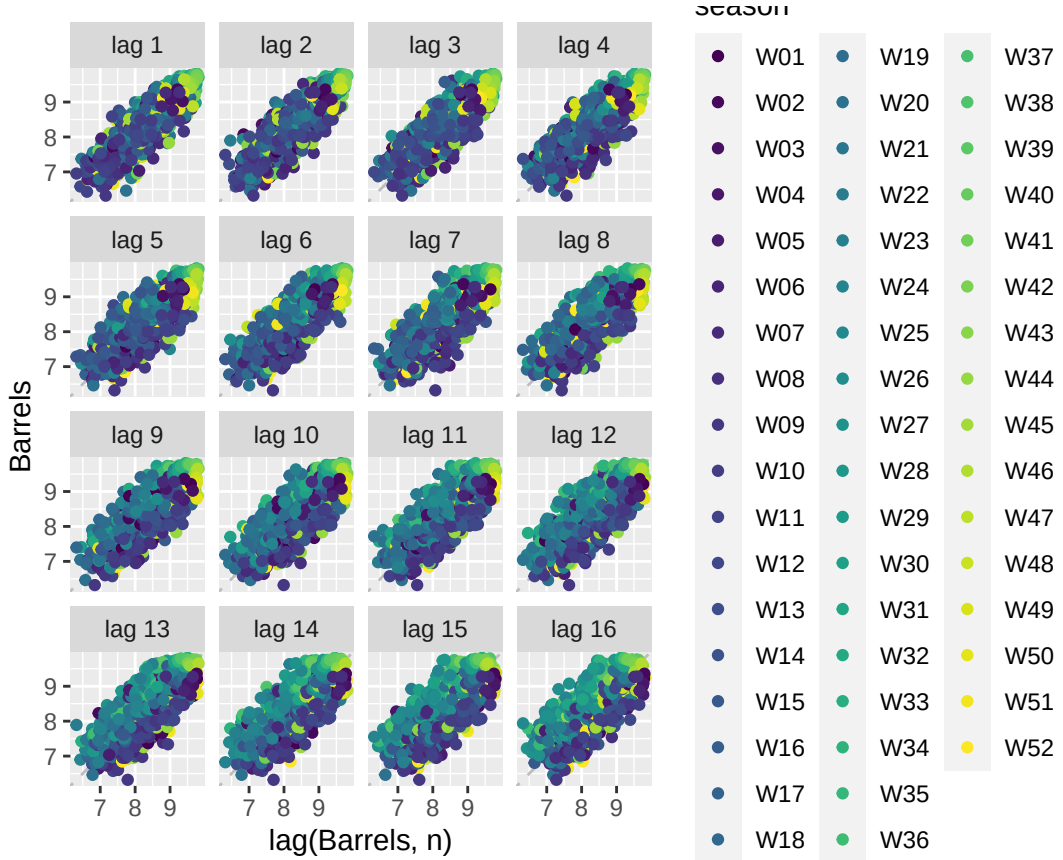

```r
# A positive trend until 2008, and then the global financial crisis led to a drop
# in sales until 2012. The shape of the seasonality seems to have changed over time.

us_gasoline %>%
  gg_season(Barrels)
```



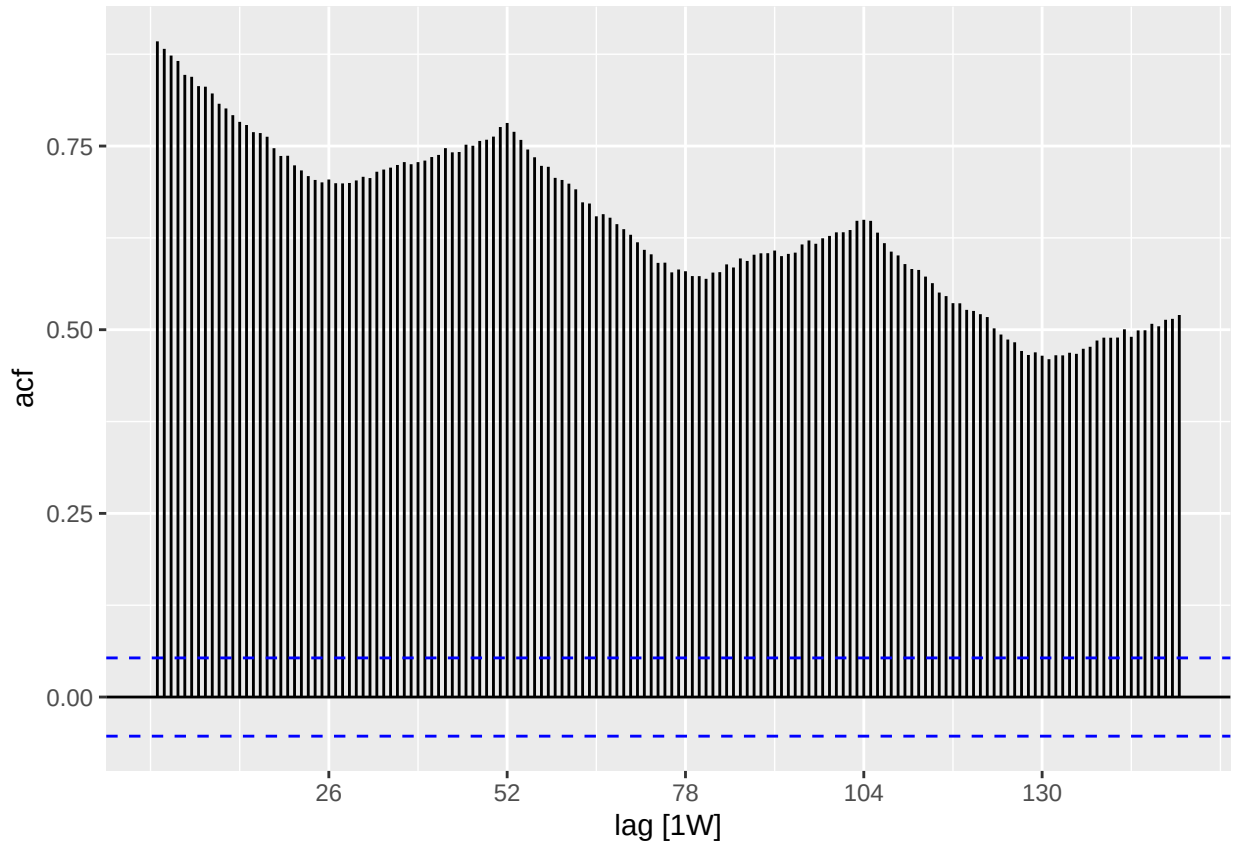

```
# There is a lot of noise making it hard to see the overall seasonal pattern.
# However, it seems to drop towards the end of quarter 4.

us_gasoline %>%
  gg_subseries(Barrels)
```




```
# The blue lines are helpful in seeing the average seasonal pattern.

us_gasoline %>%
  gg_lag(Barrels, geom='point', lags=1:16)
```




```
us_gasoline %>%
  ACF(Barrels, lag_max = 150) %>% autoplot()
```




```
# The seasonality is seen if we increase the lags to at least 2 years (approx 104 weeks)
```

3.9

Figures 3.19 and 3.20 show the result of decomposing the number of persons in the civilian labour force in Australia each month from February 1978 to August 1995.

a.

Write about 3–5 sentences describing the results of the decomposition. Pay particular attention to the scales of the graphs in making your interpretation.

```
# The Australian labour force has been decomposed into 3 components
# trend, seasonality, and remainder) using an STL decomposition.

# The trend element has been captured well by the decomposition, as it smoothly
# increases with a similar pattern to the data. The trend is of the same scale
# as the data (indicated by similarly sized grey bars), and contributes most to
# the decomposition (having the smallest scale bar).

# The seasonal component changes slowly throughout the series, with the second
# seasonal peak diminishing as time goes on - this component is the smallest
# contribution original data (having the largest scale bar).
```

```r
# The remainder is well-behaved until 1991/1992 when there is a sharp drop.
# There also appears to be a smaller drop in 1993/1994. There is sometimes some
# leakage of the trend into the remainder component when the trend window
# is too large. This appears to have happened here. It would be better if the
# recession of 1991-1992, and the smaller dip in 1993, were both included in
# the trend estimate rather than the remainder estimate. This would require a
# smaller trend window than what was used.

# In the bottom graph, the seasonal component is shown using a sub-series plot.
# December is the highest employment month, followed by March and September.
# The seasonal component changes mostly in March (with a decrease in the most
# recent years). July and August are the months with the next largest changes.
# The least changing is June with the rest are somewhere between these. December
# and September show increases in the most recent years.
```

b.

Is the recession of 1991/1992 visible in the estimated components?

```r
#Yes. The remainder shows a substantial drop during 1991 and 1992 coinciding
# with the recession.
```

5.10

a.

Create a training set for Australian takeaway food turnover (aus_retail) by withholding the last four years as a test set.

```r
takeaway <- aus_retail %>%
  filter(Industry == "Takeaway food services") %>%
  summarise(Turnover = sum(Turnover))
train <- takeaway %>%
  filter(Month <= max(Month) - 4 * 12)
```

b.

Fit all the appropriate benchmark methods to the training set and forecast the periods covered by the test set.

```r
fit <- train %>%
  model(
    naive = NAIVE(Turnover),
    drift = RW(Turnover ~ drift()),
    mean = MEAN(Turnover),
    snaive = SNAIVE(Turnover)
  )
fc <- fit %>% forecast(h = "4 years")
```

c.

Compute the accuracy of your forecasts. Which method does best?

```
fc %>%
  accuracy(takeaway) %>%
  arrange(MASE)
```

```
## # A tibble: 4 x 10
##   .model .type    ME  RMSE   MAE   MPE  MAPE  MASE RMSSE  ACF1
##   <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 naive  Test  -12.4  119.  96.4 -1.49  6.66  2.30  2.25 0.613
## 2 drift  Test  -93.7  130. 108.  -6.82  7.67  2.58  2.46 0.403
## 3 snaive Test  177.   192. 177.  11.7  11.7   4.22  3.64 0.902
## 4 mean   Test  829.   838. 829.  55.7  55.7  19.8  15.8  0.613
```

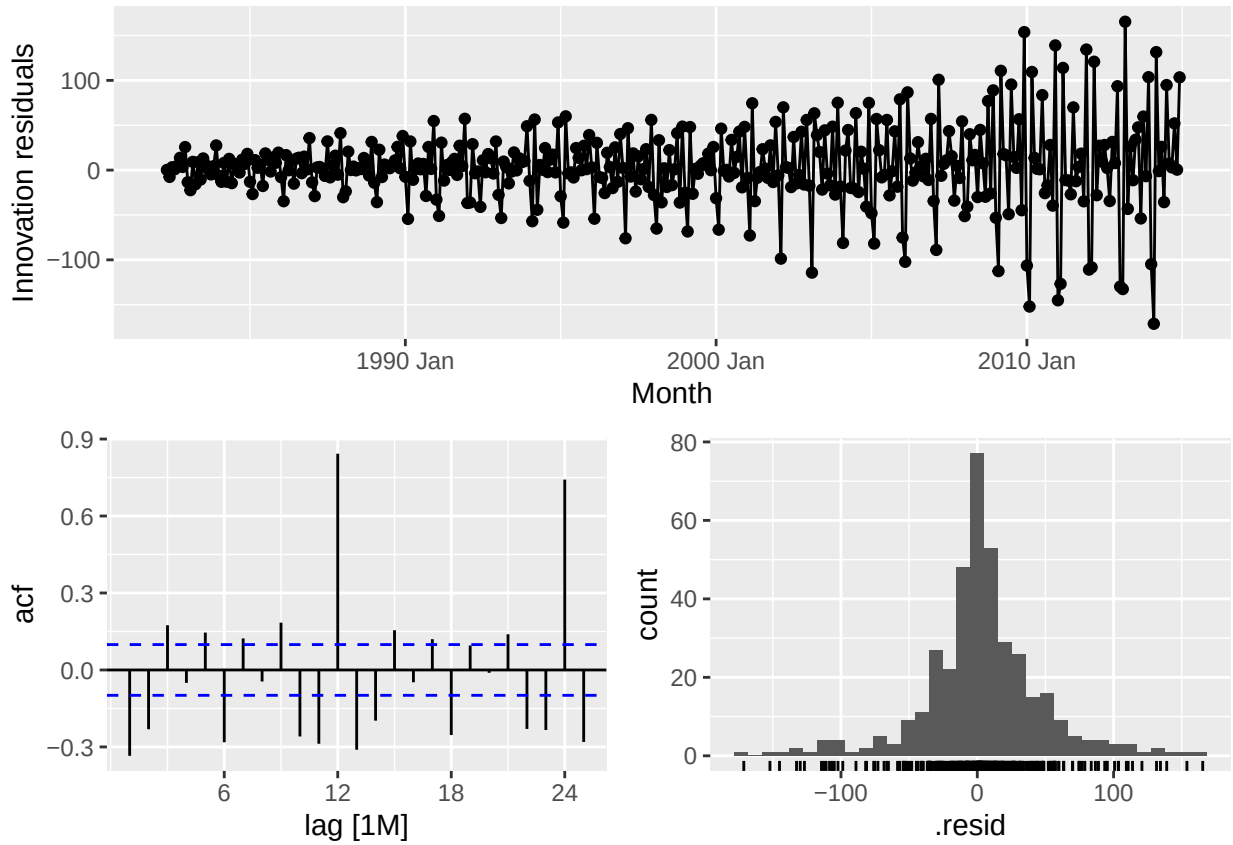
```
# The naive method is best here.
```

d.

Do the residuals from the best method resemble white noise?

```
fit %>%
  select(naive) %>%
  gg_tsresiduals()
```

```
## Warning: Removed 1 row containing missing values (`geom_line()`).

## Warning: Removed 1 rows containing missing values (`geom_point()`).

## Warning: Removed 1 rows containing non-finite values (`stat_bin()`).
```




```
# This is far from white noise. There is strong seasonality and increasing
# variance that has not been accounted for by the naive model.
```